

Ranking U.S. States by Opportunity for Single Mothers

Data sources and metrics

The analysis used publicly available datasets from **World Population Review (WPR)**. We accessed WPR's pages and their underlying view-source code to extract numeric values for each state. Each page embeds the data in a JavaScript array or `dataLookup` dictionary, which we captured and parsed. Key data sources were:

Metric	Source	Details
Cost of living index (2024)	WPR's <i>Cost of Living Index by State</i> page	The view-source page contains a <code>dataLookup</code> dictionary with an index for each state; lower values indicate more affordable living [331457298459255†L2-L7].
Childcare costs	WPR's <i>Childcare Costs by State</i> page	A <code>dataLookup</code> dictionary lists average annual cost of infant care for each state (USD) [384555122092294†L1-L2].
Crime rate	WPR's <i>Crime Rate by State 2023</i> page	The <code>data</code> array provides the 2023 total crime rate per 100 000 residents by state [945115259914303†L1-L4]. A lower crime rate suggests greater safety.
Unemployment rate	WPR's <i>Unemployment Rate by State 2023/24</i> page	A <code>dataLookup</code> dictionary contains the latest seasonally adjusted unemployment rate for each state [588224730000959†L1-L3]. Lower unemployment is better.
Public school ranking	WPR's <i>Public School Ranking by State</i> page	WPR ranks each state's K-12 public school system; the view-source contains a dictionary mapping states to ordinal ranks [602341583228605†L1-L1]. A lower rank number indicates higher quality.
Median household income (3-person households)	WPR's <i>Median Household Income</i> page	We used incomes for three-person households to approximate single-mother families. The data array lists the annual median income for each state [818151638274435†L1-L4]. Higher income improves scores.
Uninsured rate	WPR's <i>Uninsured People by State</i> page	The embedded data array lists each state's uninsured population and the percentage of residents without health insurance; example entries show <code>NoHealthInsPercentage</code> values for states [930483985155079†L168-L184]. Lower uninsured rates indicate better health care access.

Where necessary we calculated missing percentages by dividing the number of uninsured residents by the state population and multiplying by 100 (e.g., District of Columbia). We excluded the District of Columbia to focus on the 50 states.

Data cleaning and processing

Extraction: The data dictionaries were manually transcribed or copied from WPR's view-source pages. Values containing commas or currency symbols were cleaned and converted to numeric types. For instance, childcare costs like "\$7,871" were converted to `7871` and crime-rate strings were parsed as floats. Missing values (e.g., `null` uninsured percentages) were computed using population counts.

Merging: Each dataset was organized into a table keyed by state name. States appearing in all datasets were retained; District of Columbia was dropped because several metrics were unavailable.

Normalization: Metrics were normalized to make them comparable. We applied **min-max scaling** on each metric to transform values to a 0-to-1 range. For metrics where higher values indicate better conditions (median income), the normalized value was used directly. For metrics where lower values are better (cost of living, childcare cost, crime rate, unemployment rate, uninsured rate, and school ranking), the normalized score was inverted (`1 - normalized value`) so that larger scores always reflect better opportunities.

Composite scoring model

To rank states, we created a **composite score** as a weighted sum of the normalized metrics. The weights reflect the relative importance of each dimension:

Metric (normalized)	Direction	Weight
Cost-of-living index	Lower is better (inverted)	0.20
Median income	Higher is better	0.20
Childcare cost	Lower is better (inverted)	0.15
School quality rank	Lower rank number is better (inverted)	0.15
Crime rate	Lower is better (inverted)	0.10
Unemployment rate	Lower is better (inverted)	0.10
Uninsured rate	Lower is better (inverted)	0.10

These weights sum to 1.0. Affordability (cost of living, childcare, income) collectively accounts for 55% of the final score; education quality and employment each contribute 15% and 10%; safety and health care access each contribute 10%.

We calculated each state's composite score by multiplying the normalized metrics by their weights and summing the results. The states were then ranked in descending order; the highest composite score corresponds to the most favorable conditions for single mothers.

Results and interpretation

The analysis produced a ranking of all 50 states. **New Hampshire** achieved the top score because it combines high median income, low unemployment, excellent public schools, low crime, and a moderate cost of living. **North Dakota** and **New Jersey** follow closely, benefiting from strong labor markets, high incomes, and relatively low childcare costs. At the other end of the spectrum, states like **Mississippi**, **Nevada**, and **Oklahoma** ranked lower due to higher crime rates, low median incomes, or high uninsured rates.

These results should be interpreted cautiously. Data comes from 2023–2024 estimates and may not capture recent economic changes. The composite model reflects one possible weighting scheme; different weight choices (e.g., emphasizing health care or employment more strongly) would alter rankings. Additionally, the median income metric approximates the income of single-mother households but does not distinguish between family structures or account for assistance programs. Nonetheless, the analysis synthesizes multiple socioeconomic factors into a transparent ranking that can help single mothers compare opportunities across states.